Step 2 — Saved State and a Human Pause Before Anything Goes Out — Solution

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Story — Step 2: a graph can do something a hand-written loop can't: **stop in the middle, save everything, and continue later** — even after the program was closed. That's what lets a person approve an answer before it goes out. This step adds the Approval agent's file and node — `approval_node`, which for now asks a person about **every** answer — plus a checkpointer and a thread id per run. **If not:** Step 5 would have to build the pause, the saved state and the node at the same time as its checks.

After this step every answer waits for your yes/no, and a paused run survives a restart. `answer_node` and `clean_task` don't change. Files not shown here don't change.

Solution — the Approval agent's node, a SQLite checkpointer, and a real restart test

Changed from Step 1: `state.py` adds `approved`.

```
# src/state.py
# DocState: everything any node can read or write. `path` adds each node's
# entries to the list (operator.add), so it logs the nodes that ran.
# start_state(task) gives every field a starting value.
import operator
from typing import Annotated, TypedDict

class DocState(TypedDict):
    task: str
    path: Annotated[list[str], operator.add]
    # new in Step 2: whether a person (or later, the checks) approved it
    approved: bool
    answer: str

def start_state(task: str) -> DocState:
    return {"task": task, "path": [], "approved": False, "answer": ""}
```

Story — `agents/approval_agent.py`: the Approval agent's file, with its node from day one. `approval_node` calls `interrupt()`, which pauses the graph and saves the state; whatever you pass to `Command(resume=...)` comes back as `decision` (Doc09). For now its list of `reasons` is one fixed line, so **every** answer goes to a person — Step 5 replaces that one line with real checks. (Also create an empty `agents/__init__.py`.) **If not:** answers would go out with no human check at all.

```
# src/agents/approval_agent.py
# The Approval agent: approval_node() pauses with interrupt() and waits for
# a person's yes or no. On resume, the node runs again from its first line,
# so nothing above interrupt() may do a real action.
from langgraph.types import interrupt

from src.state import DocState
from src.utils.logger import get_logger

def approval_node(state: DocState) -> dict:
    logger = get_logger(__name__)
    reasons = ["every answer is checked by a person for now"]

    logger.info(f"asking a human, failed checks: {reasons}")
    decision = interrupt({"answer": state["answer"], "failed_checks": reasons})
    if decision == "yes":
        return {"approved": True, "path": ["human_approved"]}
    return {"approved": False,
            "answer": "Not approved — nothing was sent.",
            "path": ["human_rejected"]}
```

Changed from Step 1: a third node, and `build_graph()` now takes a checkpointer.

Story — `graph.py`: the checkpointer is **passed in** (Doc09's Build Task): `main.py` passes a SQLite one that writes to disk, tests pass a `MemorySaver`. **If not:** the checkpointer would be hard-coded, and a test and the app couldn't each use their own.

```
# src/graph.py
# build_graph(): adds the nodes and edges and compiles the graph. The
# checkpointer is passed in, so the app and the tests can each choose one.
from langgraph.checkpoint.memory import MemorySaver
from langgraph.graph import END, START, StateGraph

from src.agents.approval_agent import approval_node
from src.nodes import answer_node, clean_task
from src.state import DocState

def build_graph(checkpointer=None):
```

```
    if checkpointer is None:
        checkpointer = MemorySaver()

    builder = StateGraph(DocState)
    builder.add_node("clean_task", clean_task)
    builder.add_node("answer_node", answer_node)
    builder.add_node("approval_node", approval_node)

    builder.add_edge(START, "clean_task")
    builder.add_edge("clean_task", "answer_node")
    builder.add_edge("answer_node", "approval_node")
    builder.add_edge("approval_node", END)
    return builder.compile(checkpointer=checkpointer)
```

Changed from Step 1: `main.py` opens a `SqliteSaver`, gives each question its own thread id, and handles the pause with `ask_human()`.

Story — `main.py`: `SqliteSaver.from_conn_string(DB_PATH)` stores every checkpoint in `checkpoints.sqlite` (add it to `.gitignore`). Each question gets its own `thread_id`, so runs never mix. When the result contains `"__interrupt__"`, `ask_human()` shows the answer and the reason, asks yes/no, and the app resumes with `Command(resume=decision)`. **If not:** a paused run would be lost on restart, and two questions could share one saved state.

```
# main.py
# The file you run: python main.py
# Loads the settings, builds the graph, then answers questions in a loop,
# printing the path each run took.
import sys
import time

import openai
from langgraph.checkpoint.sqlite import SqliteSaver
from langgraph.types import Command

from src.config import load_config
from src.exceptions import MissingConfigError
from src.graph import build_graph
from src.state import start_state
from src.utils.logger import get_logger

DB_PATH = "checkpoints.sqlite"

def ask_human(request: dict) -> str:
    print("  -- needs your approval --")
    print(f"  answer: {request['answer']}")
    print(f"  why:   {', '.join(request['failed_checks'])}")
```

```
    return input("  Approve? (yes/no): ").strip().lower()

def main() -> None:
    try:
        load_config()
    except MissingConfigError as e:
        print(f"Setup problem: {e}")
        sys.exit(1)

    logger = get_logger(__name__)
    print("DocuMind — ask a question, or /quit to exit.")

    with SqliteSaver.from_conn_string(DB_PATH) as checkpointer:
        graph = build_graph(checkpointer)
        run_number = 0
        while True:
            task = input("You: ").strip()
            if task == "/quit":
                break
            if task == "":
                continue

            run_number = run_number + 1
            thread_id = f"run-{int(time.time())}-{run_number}"
            config = {"configurable": {"thread_id": thread_id}}
            try:
                result = graph.invoke(start_state(task), config)
                if "__interrupt__" in result:
                    request = result["__interrupt__"][0].value
                    decision = ask_human(request)
                    result = graph.invoke(Command(resume=decision), config)
            except openai.AuthenticationError:
                logger.error("Authentication failed — check OPENAI_API_KEY.")
                print("Login failed — check OPENAI_API_KEY in your .env file.")
                sys.exit(1)
            except openai.RateLimitError:
                print("We're busy right now — wait a moment and try again.")
                continue

            print(f"  path: {' -> '.join(result['path'])}")
            print(f"Answer: {result['answer']}")

if __name__ == "__main__":
    main()
```

Story — `scripts/pause_demo.py` + `scripts/resume_demo.py`: the real proof. Part 1 starts a run and stops at the pause, then exits. Part 2 is a **separate process**: it finds the paused run in the

SQLite file with `graph.get_state(config)` and finishes it. (Doc09 showed `MemorySaver` fails exactly this test.) **If not:** "it resumes" might only mean "the objects were still in memory".

```
# scripts/pause_demo.py
# Part 1 of the restart proof: starts a run that needs approval, saves the
# pause to checkpoints.sqlite, and exits.
# Run from the repo root: python -m scripts.pause_demo
from langgraph.checkpoint.sqlite import SqliteSaver

from src.config import load_config
from src.graph import build_graph
from src.state import start_state

DB_PATH = "checkpoints.sqlite"
THREAD_ID = "demo-1"
TASK = "What is the refund policy? Send it to the customer."

def main() -> None:
    load_config()
    config = {"configurable": {"thread_id": THREAD_ID}}
    with SqliteSaver.from_conn_string(DB_PATH) as checkpointer:
        graph = build_graph(checkpointer)
        result = graph.invoke(start_state(TASK), config)
    if "__interrupt__" in result:
        request = result["__interrupt__"][0].value
        print(f"Paused on thread {THREAD_ID}, waiting for approval:")
        print(f"  answer: {request['answer']}")
        print("Now close this and run: python -m scripts.resume_demo")
    else:
        print("The run finished without pausing — nothing to resume.")

if __name__ == "__main__":
    main()
```

```
# scripts/resume_demo.py
# Part 2: a brand-new process finds the paused run in checkpoints.sqlite
# with get_state() and finishes it.
# Run from the repo root: python -m scripts.resume_demo
from langgraph.checkpoint.sqlite import SqliteSaver
from langgraph.types import Command

from src.config import load_config
from src.graph import build_graph

DB_PATH = "checkpoints.sqlite"
THREAD_ID = "demo-1"
```

```
def main() -> None:
    load_config()
    config = {"configurable": {"thread_id": THREAD_ID}}
    with SqliteSaver.from_conn_string(DB_PATH) as checkpointer:
        graph = build_graph(checkpointer)
        saved = graph.get_state(config)
        if not saved.next:
            print(f"Nothing is paused on thread {THREAD_ID}.")
            return
        print(f"Found a paused run: {saved.values['task']}")
        decision = input("Approve? (yes/no): ").strip().lower()
        result = graph.invoke(Command(resume=decision), config)
    print(f"path: {' -> '.join(result['path'])}")
    print(f"Answer: {result['answer']}")

if __name__ == "__main__":
    main()
```

Changed from Step 1: `tests/integration/test_live_graph.py` gets cases 2-4. `run()` gains one optional parameter, `decision`: if the run paused, it resumes with that answer (`None` means "leave it paused"). The test builds the graph with a fresh `MemorySaver`, so nothing is written to disk.

Story — `tests/integration/test_live_graph.py`: the pause is this step's feature, so it gets three checks: an action task pauses, "yes" approves it, and "no" stops it. **If not:** a pause that didn't stop anything — or a "no" that still sent the answer — would go unnoticed.

```
# tests/integration/test_live_graph.py
# Runs the whole graph against the real OpenAI API — each step adds cases.
# Each case prints OK or WRONG. Needs a valid OPENAI_API_KEY in .env.
# Run from the repo root: python -m tests.integration.test_live_graph
# new in Step 2
from langgraph.checkpoint.memory import MemorySaver
from langgraph.types import Command

from src.config import load_config
from src.graph import build_graph
from src.state import start_state

# new in Step 2
ACTION_TASK = "What is the refund policy? Send it to the customer."


# changed in Step 2: resumes a paused run with `decision` (None = don't)
def run(
    graph, thread_id: str, task: str, decision: str | None = "yes"
```

```
) -> dict:
    config = {"configurable": {"thread_id": thread_id}}
    outcome = graph.invoke(start_state(task), config)
    if decision is not None and "__interrupt__" in outcome:
        outcome = graph.invoke(Command(resume=decision), config)
    return outcome


def show(case: str, passed: bool, outcome: dict) -> None:
    result = "OK"
    if not passed:
        result = "WRONG"
    print(f"{case} -> {result}: {outcome.get('path')}")


def test_general_question(graph) -> None:
    outcome = run(graph, "t1", "What is the capital of France?")
    passed = "Paris" in outcome["answer"] and "answer_node" in outcome["path"]
    show("case 1 (general question)", passed, outcome)


# new in Step 2
def test_action_pauses(graph) -> None:
    outcome = run(graph, "t2", ACTION_TASK, decision=None)
    show("case 2 (pauses)          ", "__interrupt__" in outcome, outcome)


# new in Step 2
def test_human_says_yes(graph) -> None:
    outcome = run(graph, "t3", ACTION_TASK, "yes")
    passed = outcome["approved"] and "human_approved" in outcome["path"]
    show("case 3 (human: yes)      ", passed, outcome)


# new in Step 2
def test_human_says_no(graph) -> None:
    outcome = run(graph, "t4", ACTION_TASK, "no")
    passed = outcome["answer"].startswith("Not approved")
    show("case 4 (human: no)       ", passed, outcome)


if __name__ == "__main__":
    load_config()
    # changed in Step 2: a fresh MemorySaver, nothing on disk
    graph = build_graph(MemorySaver())
    test_general_question(graph)
    # new in Step 2
```

```
    test_action_pauses(graph)
    test_human_says_yes(graph)
    test_human_says_no(graph)
```

Expected output of `python -m tests.integration.test_live_graph` (every answer pauses in this step, and case 1 is resumed with "yes"):

```
case 1 (general question) -> OK: ['clean_task', 'answer_node',
'human_approved']
case 2 (pauses)          -> OK: ['clean_task', 'answer_node']
case 3 (human: yes)      -> OK: ['clean_task', 'answer_node',
'human_approved']
case 4 (human: no)       -> OK: ['clean_task', 'answer_node',
'human_rejected']
```

(Wrapped to fit the page.)

Run it:

```
python -m scripts.pause_demo      # pauses, then exits
python -m scripts.resume_demo     # a new process finishes the run
python main.py
```

Expected output (the answer's wording varies):

```
$ python -m scripts.pause_demo
Paused on thread demo-1, waiting for approval:
  answer: Refund policies usually allow returns within 14-30 days ...
Now close this and run: python -m scripts.resume_demo

$ python -m scripts.resume_demo
Found a paused run: What is the refund policy? Send it to the customer.
Approve? (yes/no): yes
path: clean_task -> answer_node -> human_approved
Answer: Refund policies usually allow returns within 14-30 days ...
```

Notice the answer is a *general* guess — the graph can't read your documents yet. That's Step 3.